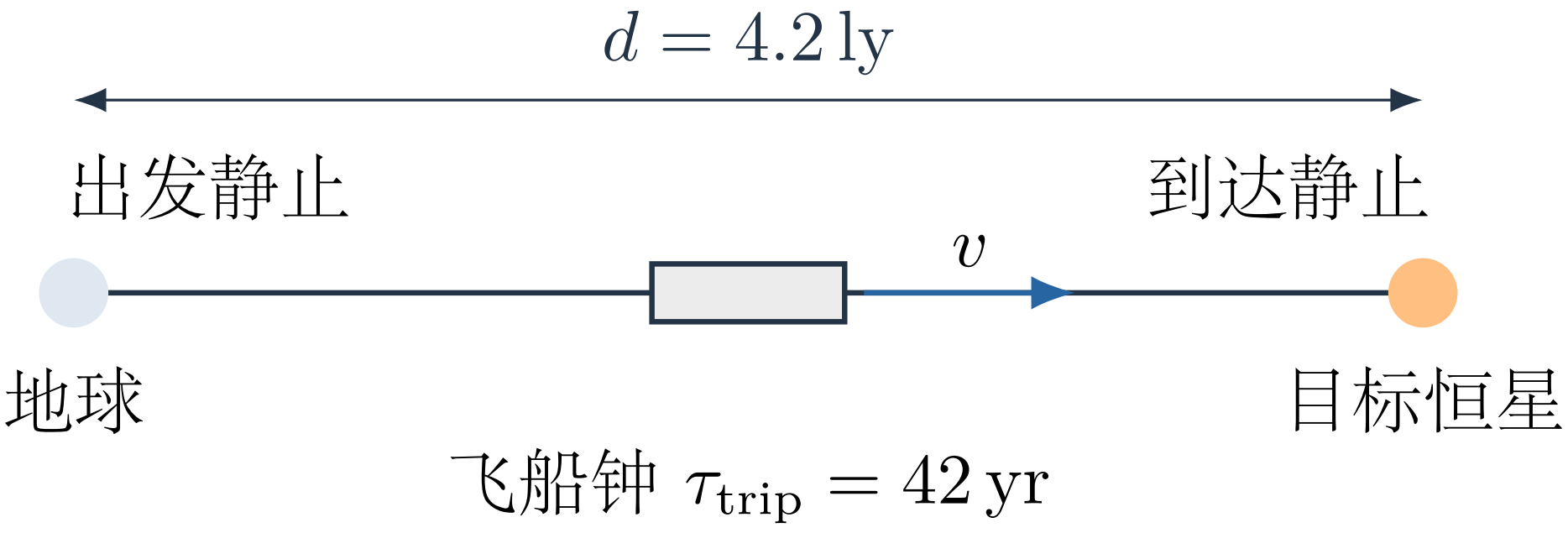
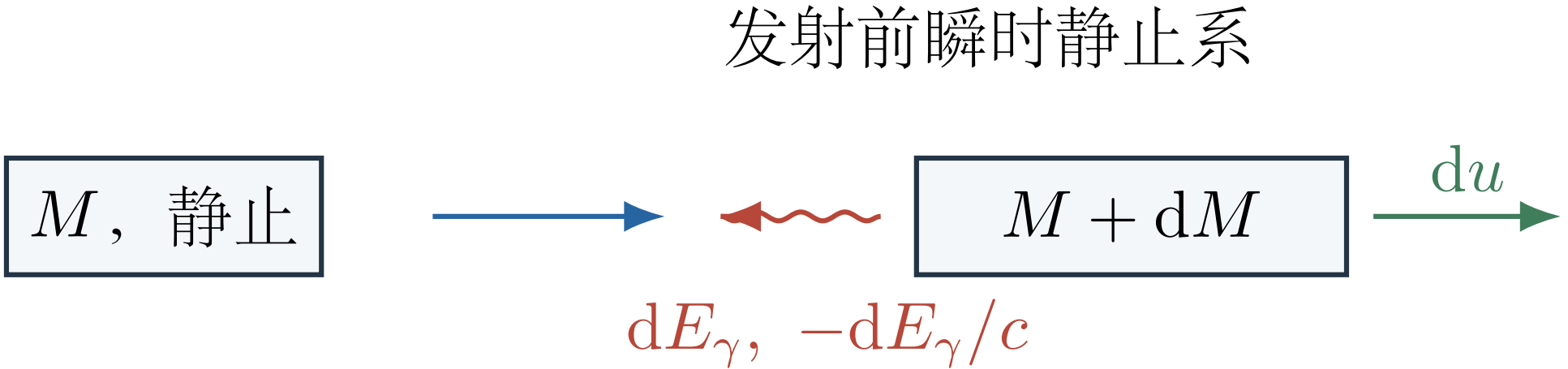
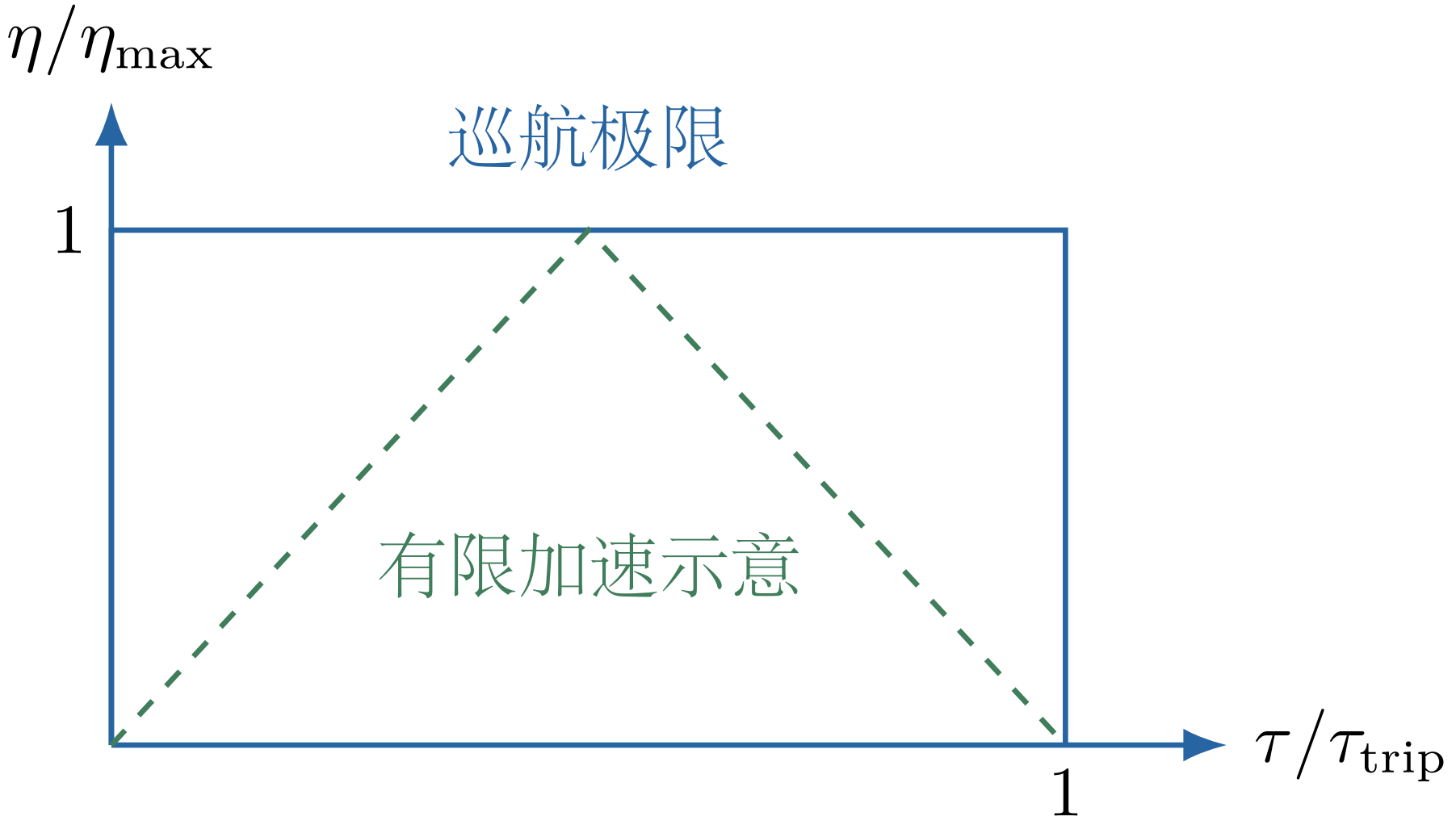
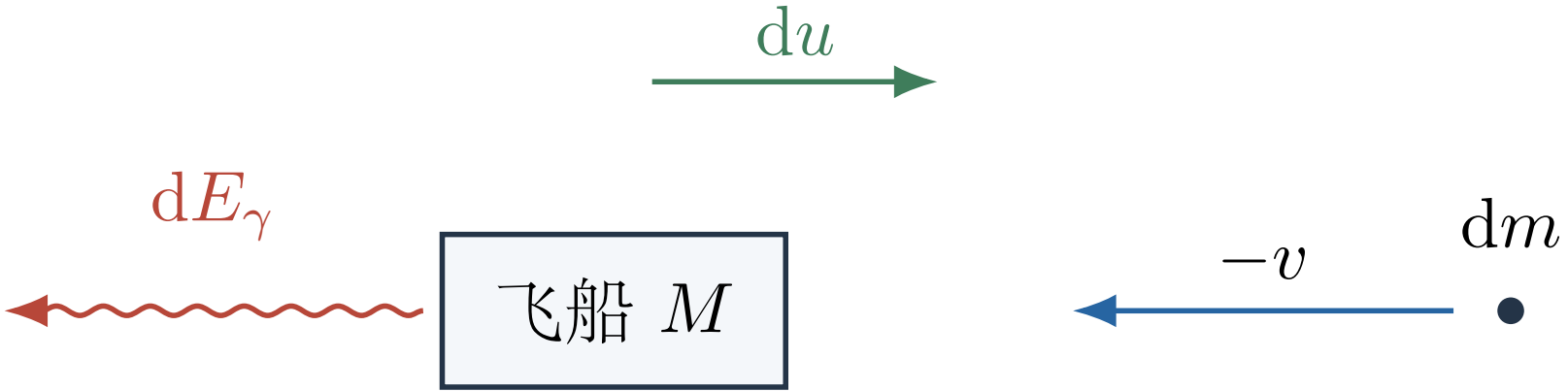

$d = 4.2\,\mathrm{ly}$
出发静止
到达静止
v
地球
目标恒星
飞船钟 $\tau_{\mathrm{trip}} = 42\,\mathrm{yr}$


发射前瞬时静止系
M，静止
$M+\mathrm{d}M$
$\mathrm{d}u$
$\mathrm{d}E_\gamma,\ -\mathrm{d}E_\gamma/c$


$\eta/\eta_{\max}$
巡航极限
1
有限加速示意
1
τ/τ_{trip}



消耗正物质 $\mathrm{d}m$；剩余静质量 $M - \mathrm{d}m$